RAJALAKSHMI ENGINEERING COLLEGE

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C2

Started on	Thursday, 17 September 2026, 8:23 AM
State	Finished
Completed on	Thursday, 17 September 2026, 8:23 AM
Time taken	36 secs
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 10.00 out of 10.00

Playing with Chessboard:

Ram is given with an n*n chessboard with each cell with a monetary value. Ram stands at the (0,0), that the position of the top left white rook. He is been given a task to reach the bottom right black rook position (n-1, n-1) constrained that he needs to reach the position by traveling the maximum monetary path under the condition that he can only travel one step right or one step down the board. Help ram to achieve it by providing an efficient DP algorithm.

Example:

Input

3

1 2 4

2 3 4

8 7 1

Output:

19

Explanation:

Totally there will be 6 paths among that the optimal is

Optimal path value:1+2+8+7+1=19

Input Format

First Line contains the integer n

The next n lines contain the n*n chessboard values

Output Format

Print Maximum monetary value of the path

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

#define MAX_N 500 // Adjust if n can be larger

int matrix[MAX_N][MAX_N];
int dp[MAX_N][MAX_N];

int main() {
    int n;

    // Read the grid size
    if (scanf("%d", &n) != 1) return 0;

    // Read the chessboard cell monetary values
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n; j++) {
            if (scanf("%d", &matrix[i][j]) != 1) return 0;
        }
    }

    // Initialize the starting position
    dp[0][0] = matrix[0][0];

    // Initialize the first row (can only come from the left cell)
    for (int j = 1; j < n; j++) {
        dp[0][j] = dp[0][j - 1] + matrix[0][j];
    }

    // Initialize the first column (can only come from the top cell)
    for (int i = 1; i < n; i++) {
        dp[i][0] = dp[i - 1][0] + matrix[i][0];
    }

    // Fill the rest of the DP table
    for (int i = 1; i < n; i++) {
        for (int j = 1; j < n; j++) {
            int top = dp[i - 1][j];
            int left = dp[i][j - 1];
```

```
            int left = dp[i][j - 1];

            // Choose the path that gives the maximum monetary value
            if (top > left) {
                dp[i][j] = matrix[i][j] + top;
            } else {
                dp[i][j] = matrix[i][j] + left;
            }
        }
    }

    // The answer is at the bottom-right corner of the board
    printf("%d\n", dp[n - 1][n - 1]);

    return 0;
```

	Input	Expected	Got	
✔	3 1 2 4 2 3 4 8 7 1	19	19	✔
✔	3 1 3 1 1 5 1 4 2 1	12	12	✔
✔	4 1 1 3 4 1 5 7 8 2 3 4 6 1 6 9 0	28	28	✔

Passed all tests! ✔

Correct

Marks for this submission: 10.00/10.00.

Back to Course